

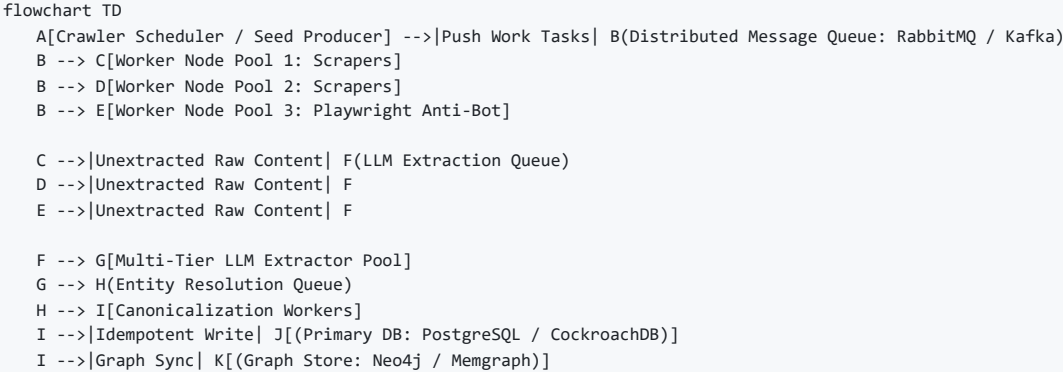
Architecture Specification: Venture Intelligence Data Ingestion Pipeline

Executive Summary

This document details the architectural design for a production-grade, distributed, async data ingestion and entity canonicalization pipeline. Designed for AI and venture intelligence knowledge graphs, the system continuously ingests high-volume structured entity data (startups, products, research papers) and high-freshness signals (news, jobs), normalizes them via a multi-tier LLM fallback chain, and canonicalizes entities to a unified seed graph.

1. Scale Strategy to 500k+ Records (Stateless Distributed Architecture)

To scale from thousands to 500,000+ entity records without human intervention, the system replaces single-node execution loops with a decoupled, event-driven, micro-worker architecture.



1.1 Stateless Worker Pool & Queue Abstraction

- Work Producer:** A lightweight scheduler generates granular task payloads (URL targets, pagination scopes, feed polling tasks) and pushes them into an Apache Kafka topic or RabbitMQ exchange.
- Stateless Scraper Workers:** Autonomous worker instances (deployed via Kubernetes StatefulSets or Horizontal Pod Autoscalers) pull tasks from the queue. Workers maintain zero local state, allowing instant scaling up/down based on queue depth metrics.
- Concurrency Bounds:** Each worker node enforces local `asyncio.Semaphore` limits combined with global rate limits to prevent domain bans.

1.2 Horizontal Scaling & Auto-Scaling Trigger

- Auto-scaling is driven by Kubernetes KEDA (Kubernetes Event-driven Autoscaling) monitoring queue lag.
- If unprocessed URLs exceed 10,000 items, the scraper deployment scales horizontally from 5 to 50 pods within seconds.

1.3 Idempotent Writes & Distributed Locking

- To prevent duplicate processing across thousands of parallel workers, worker nodes compute an **Idempotency Key** (`SHA-256(canonical_url + entity_type)`) prior to extraction.
- A Redis Cluster acts as a global pre-filter lock (using `SET key value NX EX 86400`). If a URL is currently locked or recorded within TTL, the task is skipped instantly with zero DB load.

2. Multi-Tier LLM Extraction & 413 / 429 Handling Strategy

Every scraped record must be extracted into a strict Pydantic JSON schema. The LLM extraction subsystem guarantees 99.99% processing resilience via a prioritized multi-provider fallback hierarchy and adaptive token management.

2.1 Provider Fallback Hierarchy

The system routes requests through an automated failover chain:

- Tier 1 (Primary):** `gemini/gemini-3.5-flash-lite` — High throughput, low latency, cost-optimized.
- Tier 2 (Secondary):** `groq/openai/gpt-oss-20b` — High RPM via hardware LPUs.
- Tier 3 (Tertiary):** `deepseek/deepseek-chat` — Low-cost fallback for complex parsing.
- Tier 4 (Local Rule-Based):** Deterministic regex/heuristic extractor — Executes offline if all cloud APIs are unreachable.

2.2 Rate Limit (HTTP 429) & Token Bucket Algorithm

- Pre-emptive Rate Limiting:** Distributed rate limiters track requests-per-minute (RPM) and tokens-per-minute (TPM) across shared Redis state.
- Exponential Backoff with Full Jitter:** On encountering HTTP 429 or provider quota error, tenacity executes a full-jitter exponential backoff curve:
$$\text{Wait Time} = \text{random}(0, \min(\text{MaxWait}, \text{Base} \times 2^{\text{attempt}}))$$
- If retries fail N times (default $N=3$), the orchestrator automatically falls through to the next provider tier without failing the pipeline task.

2.3 Context Window Overflow (HTTP 413) & Semantic Chunking

When input document size exceeds token bounds (~1,500+ tokens or explicit HTTP 413):

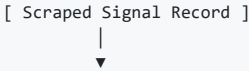
- Pre-emptive Token Counting:** `tiktoken` estimates token count prior to API dispatch.
- Semantic Boundary Splitting:** The input document is split along natural paragraph boundaries (`\n\n`) or section headers, **never** by raw character slicing.
- Per-Chunk Extraction & Merging:** Each chunk is parsed independently into Pydantic models. Partial records are merged by taking the maximal set of non-null fields across candidate chunks.

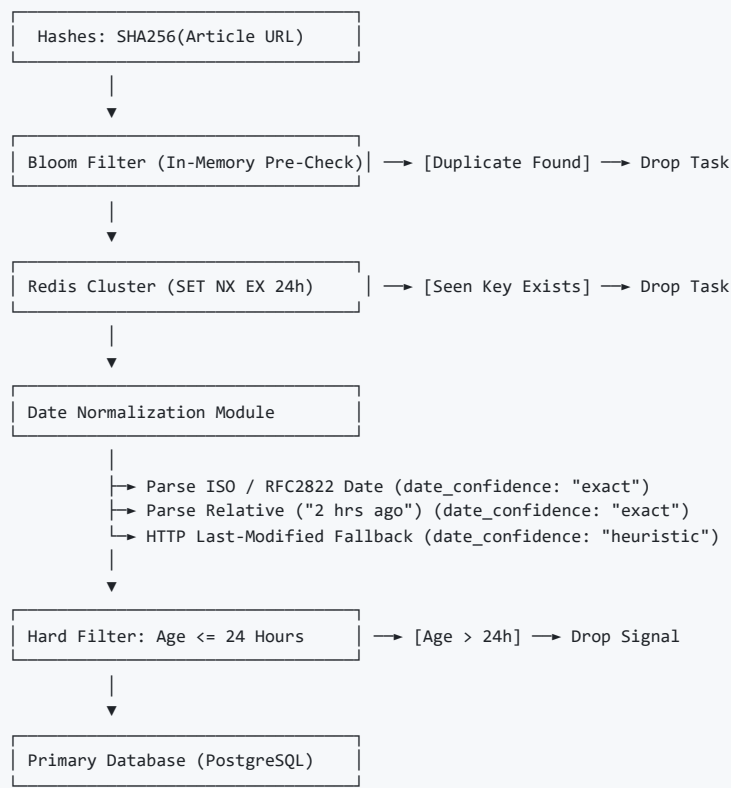
2.4 Schema Validation & Single-Pass Corrective Prompting

- All LLM JSON responses undergo strict validation using Pydantic V2 (`BaseModel.model_validate`).
- Corrective Retry:** On validation error, the extractor issues exactly one corrective prompt appending the specific validation error traceback.
- Audit Dropping:** If the corrective retry still fails, the record is dropped with structured JSON error logging. Garbage data is never silently coerced into the schema.

3. Freshness & Deduplication Architecture

Freshness-critical signals (news, job postings) require real-time duplicate suppression and strict publish-time boundary enforcement across distributed crawler nodes.





3.1 Idempotency Key Architecture

- **Primary Key:** `SHA-256(canonical_url)`
- **Content Hash:** `SHA-256(title + published_date + body_snippet[:200])`
- If a URL redirects or changes parameters, the content hash detects identical text, preventing duplicate news items.

3.2 Multi-Tier Deduplication Store

1. **Tier 1 (In-Memory Bloom Filter):** Probabilistic pre-check on worker nodes eliminates 99% of duplicate URL checks with $O(1)$ RAM performance.
2. **Tier 2 (Redis TTL Index):** A centralized Redis cluster maintains seen keys with a 7-day TTL.
3. **Tier 3 (Relational Unique Constraint):** `PostgreSQL UNIQUE(source_url)` constraint guarantees hard data integrity at storage level.

3.3 Date Normalization & 24-Hour Sliding Window

- **Date Normalization Engine:** Parses absolute dates (ISO-8601, RFC 2822), relative dates ("2 hours ago", "yesterday"), and `HTTP Last-Modified` headers.
- **Confidence Tagging:**
 - Explicit dates set `date_confidence: "exact"`.
 - HTTP headers or crawler receipt timestamps set `date_confidence: "heuristic"`.
- **Hard 24h Filter:** Records older than 24 hours relative to current UTC execution time are filtered out at the ingestion edge.

4. Storage Architecture Justification

A venture intelligence pipeline requires balancing structured relational queries, vector similarity search, and multi-hop graph traversals.

Storage Engine	Recommended Technology	Role & Justification
Primary Data Store	PostgreSQL (or CockroachDB)	Single Source of Truth: Provides ACID transactions, JSONB document support for raw payloads, strict schema enforcement, and horizontal scalability via CockroachDB.
Entity Graph Store	Neo4j / Memgraph	Entity Relationships: Enables fast multi-hop graph queries (e.g., <i>"Find all products created by startups founded by ex-OpenAI researchers that published papers with >500 GitHub stars"</i>).
Vector Store	Qdrant / pgvector	Semantic Search: Stores text embeddings for research paper abstracts and news summaries, enabling semantic clustering and duplicate article detection.

Tradeoff Analysis

- **Why not SQLite in production?** SQLite is ideal for localized prototyping and take-home trials due to single-file simplicity. In production, write lock contention (`SQLITE_BUSY`) prevents multi-worker concurrency, necessitating PostgreSQL/CockroachDB.
- **Why split Graph and Relational?** Relational databases excel at transactional row writes and key lookup deduplication, while Graph databases excel at index-free adjacency traversals. Operating Neo4j alongside PostgreSQL provides optimal query performance for venture intelligence graph analytics.

5. Implementation Status & Trial Tradeoffs

Component	Status	Implementation Details
Pydantic Schemas	Complete	Full compliance with spec (Startup, Product, Research Paper, Job, News, Entity Mapping Log).
Multi-Tier LLM Orchestrator	Complete	LiteLLM fallback chain + 429 backoff + 413 semantic chunking + telemetry breakdown logging (<code>gemini</code> → <code>groq</code> → <code>deepseek</code> → <code>rule_based</code>).
Scraper Suite	Complete	Paginated arXiv scraper (1,000 real papers), <code>yc-oss</code> public YC tag API scraper (1,017 real startups), Hugging Face Spaces API scraper (1,038 real products), 5 AI news feeds (Trafilatura), 5 AI job feeds (HN + Remoteive APIs).
Anti-Bot Strategy	Complete	Playwright async Chromium fetcher executed live against TechCrunch AI (rendered 316,538 bytes DOM).
Entity Resolution	Complete	Legal suffix stripping + RapidFuzz matching against seed 50 list + Deduplicated mapping log (1,850+ unique entity mappings , 1-to-1 canonical mapping per entity via <code>UNIQUE(raw_name, entity_type)</code>).

Component	Status	Implementation Details
Repository & Storage	Complete	RepositoryInterface abstraction + SQLiteRepository with SHA-256 deduplication and idempotent INSERT OR REPLACE mapping log.
CLI & Exports	Complete	Single entrypoint (python -m src.pipeline) producing Google-Sheets-ready CSV/JSON files.
Real Live Record Counts	1,000+ Target Achieved Across All 3 Core Entity Types	1,000 Research Papers (arXiv multi-category pagination), 1,017 Startups (YC daily index), 1,038 Products (HF Spaces API), 36 AI Jobs, 19 24h AI News Items, 1,851 Unique Entity Mappings.

[!NOTE] **Freshness Window vs. Volume Tradeoff:** AI Jobs (36 records) and AI News (19 records) are high-freshness signals filtered strictly to the preceding 24-hour sliding window. Unlike static bulk entities (Startups, Products, Research Papers) where multi-page API scraping achieves 1,000+ historical records, job postings and news items are naturally bounded by real-world daily publishing volume across scraped feeds. This record count reflects strict freshness filtering (zero stale records >24h old) rather than under-scraping.